

Source Reference List

Every source cited while teaching Module 6 (Databases & SQL for ML) and Module 7 (Advanced Databases & SQL for ML), organized by topic and subtopic, with a note on what each source is useful for.

MODULE 6: Databases & SQL for ML

Topic 1: Database Fundamentals

1a. What is a database

https://www.w3schools.com/sql/sql_intro.asp

For: Beginner-friendly intro to what databases and SQL are

<https://www.postgresql.org/docs/current/tutorial-start.html>

For: Official conceptual grounding on databases

<https://www.freecodecamp.org/news/relational-database-basics/>

For: Article-style intro to relational database basics

1b. Why databases are important

<https://www.ibm.com/topics/database>

For: IBM's overview of why databases matter

<https://www.oracle.com/database/what-is-database/>

For: Oracle's explanation of database purpose and value

<https://www.freecodecamp.org/news/relational-database-basics/>

For: Why use a database instead of flat files

1c. Types of databases

<https://aws.amazon.com/products/databases/>

For: AWS overview of all major database types

<https://www.mongodb.com/resources/basics/databases/nosql-explained/nosql-vs-sql>

For: SQL vs NoSQL comparison

<https://cloud.google.com/learn/what-is-a-data-warehouse>

For: What a data warehouse is, as a database type

1d. Relational databases

<https://www.postgresqltutorial.com/postgresql-getting-started/postgresql-sample-database/>

For: Practical relational database concepts

<https://www.ibm.com/topics/relational-databases>

For: IBM's definition and explanation of relational databases

<https://www.oracle.com/database/what-is-a-relational-database/>

For: Oracle's explanation of relational databases

1e. Database use cases in ML

<https://cloud.google.com/vertex-ai/docs/featurestore/overview>

For: Feature store overview for ML data storage

<https://www.databricks.com/glossary/feature-store>

For: What a feature store is and why it's used in ML

<https://mlflow.org/docs/latest/tracking.html>

For: ML experiment tracking, an ML database use case

Topic 2: Relational Database Concepts

2a. Tables

https://www.w3schools.com/sql/sql_ref_table.asp

For: SQL table reference and structure

<https://www.postgresqltutorial.com/postgresql-tutorial/postgresql-create-table/>

For: How tables are created and structured

<https://www.khanacademy.org/computing/computer-programming/sql>

For: Intro to SQL tables course

2b. Rows and columns

https://www.w3schools.com/sql/sql_syntax.asp

For: Basic SQL syntax covering rows/columns

<https://mode.com/sql-tutorial/introduction-to-sql/>

For: SQL tutorial covering basic table structure

2c. Primary keys

https://www.w3schools.com/sql/sql_primarykey.ASP

For: Primary key syntax and rules

<https://www.postgresqltutorial.com/postgresql-tutorial/postgresql-primary-key/>

For: Primary key concept and examples

<https://www.geeksforgeeks.org/dbms/primary-key-in-relational-model/>

For: Primary key theory in DBMS

2d. Foreign keys

https://www.w3schools.com/sql/sql_foreignkey.asp

For: Foreign key syntax and rules

<https://www.postgresqltutorial.com/postgresql-tutorial/postgresql-foreign-key/>

For: Foreign key concept and examples

<https://www.geeksforgeeks.org/dbms/foreign-key/>

For: Foreign key theory in DBMS

2e. Relationships

<https://www.lucidchart.com/pages/database-diagram/database-relationships>

For: Visual explanation of database relationships

<https://www.postgresqltutorial.com/postgresql-tutorial/postgresql-many-to-many/>

For: Many-to-many relationships and junction tables

<https://www.geeksforgeeks.org/dbms/types-of-relationships-in-dbms/>

For: Types of relationships in DBMS

Topic 3: SQL Fundamentals

3a. SQL syntax

https://www.w3schools.com/sql/sql_syntax.asp

For: SQL syntax basics and conventions

<https://dev.mysql.com/doc/refman/8.0/en/sql-statements.html>

For: Official MySQL SQL statement syntax reference

<https://mode.com/sql-tutorial/introduction-to-sql/>

For: Intro to SQL, category overview (DDL/DML/etc.)

3b. Creating databases

<https://dev.mysql.com/doc/refman/8.0/en/create-database.html>

For: Official CREATE DATABASE statement reference

https://www.w3schools.com/mysql/mysql_create_db.asp

For: CREATE DATABASE syntax and examples

3c. Creating tables

<https://dev.mysql.com/doc/refman/8.0/en/create-table.html>

For: Official CREATE TABLE statement reference

https://www.w3schools.com/sql/sql_create_table.asp

For: CREATE TABLE syntax and examples

3d. Data types

<https://dev.mysql.com/doc/refman/8.0/en/data-types.html>

For: Official MySQL data types reference

https://www.w3schools.com/mysql/mysql_datatypes.asp

For: MySQL data types summary and examples

3e. Constraints

<https://dev.mysql.com/doc/refman/8.0/en/constraints.html>

For: Official MySQL constraints reference

https://www.w3schools.com/sql/sql_constraints.asp

For: SQL constraints syntax and examples

Topic 4: CRUD Operations

4a. INSERT

<https://dev.mysql.com/doc/refman/8.0/en/insert.html>

For: Official INSERT statement reference

https://www.w3schools.com/sql/sql_insert.asp

For: INSERT INTO syntax and examples

4b. SELECT

<https://dev.mysql.com/doc/refman/8.0/en/select.html>

For: Official SELECT statement reference

https://www.w3schools.com/sql/sql_select.asp

For: SELECT syntax and examples

4c. UPDATE

<https://dev.mysql.com/doc/refman/8.0/en/update.html>

For: Official UPDATE statement reference

https://www.w3schools.com/sql/sql_update.asp

For: UPDATE syntax and examples

4d. DELETE

<https://dev.mysql.com/doc/refman/8.0/en/delete.html>

For: Official DELETE statement reference

https://www.w3schools.com/sql/sql_delete.asp

For: DELETE syntax and examples

4e. Data management workflows

<https://developer.mozilla.org/en-US/docs/Glossary/CRUD>

For: CRUD concept explained (MDN glossary)

<https://www.freecodecamp.org/news/crud-operations-explained/>

For: CRUD operations explained end-to-end

Topic 5: Filtering & Sorting

5a. WHERE clause

<https://dev.mysql.com/doc/refman/8.0/en/where-optimization.html>

For: Official WHERE clause reference and optimization

https://www.w3schools.com/sql/sql_where.asp

For: WHERE clause syntax and examples

5b. ORDER BY

<https://dev.mysql.com/doc/refman/8.0/en/order-by-optimization.html>

For: Official ORDER BY reference and optimization

https://www.w3schools.com/sql/sql_orderby.asp

For: ORDER BY syntax and examples

5c. LIMIT

<https://dev.mysql.com/doc/refman/8.0/en/limit-optimization.html>

For: Official LIMIT clause reference

https://www.w3schools.com/sql/sql_top.asp

For: LIMIT / TOP-N row syntax across databases

5d. DISTINCT

<https://dev.mysql.com/doc/refman/8.0/en/distinct-optimization.html>

For: Official DISTINCT reference and optimization

https://www.w3schools.com/sql/sql_distinct.asp

For: DISTINCT syntax and examples

5e. Practical data retrieval

<https://dev.mysql.com/doc/refman/8.0/en/select.html>

For: Full SELECT syntax reference (clause ordering)

<https://mode.com/sql-tutorial/sql-query-order-of-operations/>

For: SQL query order of execution explained

Topic 6: Aggregate Functions

6a. COUNT

https://dev.mysql.com/doc/refman/8.0/en/aggregate-functions.html#function_count

For: Official COUNT() function reference

https://www.w3schools.com/sql/sql_count_avg_sum.asp

For: COUNT/AVG/SUM syntax and examples

6b. SUM

https://dev.mysql.com/doc/refman/8.0/en/aggregate-functions.html#function_sum

For: Official SUM() function reference

https://www.w3schools.com/sql/sql_count_avg_sum.asp

For: COUNT/AVG/SUM syntax and examples

6c. AVG

https://dev.mysql.com/doc/refman/8.0/en/aggregate-functions.html#function_avg

For: Official AVG() function reference

https://www.w3schools.com/sql/sql_count_avg_sum.asp

For: COUNT/AVG/SUM syntax and examples

6d. MIN

https://dev.mysql.com/doc/refman/8.0/en/aggregate-functions.html#function_min

For: Official MIN()/MAX() function reference

https://www.w3schools.com/sql/sql_min_max.asp

For: MIN/MAX syntax and examples

6e. MAX

https://dev.mysql.com/doc/refman/8.0/en/aggregate-functions.html#function_min

For: Official MIN()/MAX() function reference

https://www.w3schools.com/sql/sql_min_max.asp

For: MIN/MAX syntax and examples

Topic 7: ML Data Storage Concepts

7a. Dataset storage

<https://cloud.google.com/learn/what-is-a-data-lake>

For: Data lakes vs data warehouses for dataset storage

<https://aws.amazon.com/products/databases/>

For: Choosing the right data store for a workload

7b. Structured data management

<https://www.ibm.com/topics/structured-vs-unstructured-data>

For: Structured vs unstructured data explained

<https://cloud.google.com/architecture/data-validation-patterns>

For: Data validation best practices

7c. Data retrieval workflows

<https://airflow.apache.org/docs/apache-airflow/stable/core-concepts/overview.html>

For: Scheduled/automated data pipeline concepts

<https://cloud.google.com/discover/batch-processing-vs-stream-processing>

For: Batch vs streaming data retrieval

7d. Data consistency

<https://www.ibm.com/topics/data-integrity>

For: Data integrity vs data consistency explained

<https://dev.mysql.com/doc/refman/8.0/en/constraints.html>

For: How constraints enforce consistency

7e. Data quality awareness

<https://cloud.google.com/learn/what-is-data-quality>

For: What data quality is and why it matters

<https://docs.greatexpectations.io/docs/>

For: Great Expectations: a real data quality validation tool

MODULE 7: Advanced Databases & SQL for ML

Topic 1: Data Modeling

1a. Entity identification

<https://www.lucidchart.com/pages/er-diagrams>

For: Entity Relationship Diagram (ERD) guide

<https://www.ibm.com/topics/data-modeling>

For: What data modeling is, IBM overview

<https://vertabelo.com/blog/data-modeling-101/>

For: How to identify entities in data modeling

1b. Relationships

<https://vertabelo.com/blog/data-modeling-101/>

For: Entity relationships in data modeling

<https://www.lucidchart.com/pages/database-diagram/database-relationships>

For: Database relationships explained visually

1c. Data modeling principles

<https://vertabelo.com/blog/data-modeling-101/>

For: Data modeling best practices

<https://learn.microsoft.com/en-us/office/troubleshoot/access/database-design-basics>

For: Database design basics from Microsoft

1d. Real-world schema design

<https://vertabelo.com/blog/data-modeling-101/>

For: The full database design process

<https://dev.mysql.com/doc/refman/8.0/en/create-table.html>

For: Official CREATE TABLE reference for implementing schemas

1e. Scalable database structures

<https://aws.amazon.com/products/databases/>

For: Database scalability best practices

<https://vertabelo.com/blog/data-modeling-101/>

For: Scaling databases as part of good design

<https://dev.mysql.com/doc/refman/8.0/en/partitioning.html>

For: Official MySQL partitioning reference

Topic 2: Database Normalization

2a. Why normalization matters

<https://vertabelo.com/blog/normalization-101/>

For: Full normalization guide and rationale

<https://www.geeksforgeeks.org/dbms/normal-forms-in-dbms/>

For: Normal forms overview in DBMS

2b. First Normal Form (1NF)

<https://www.geeksforgeeks.org/dbms/first-normal-form-1nf/>

For: 1NF definition, rules, and examples

<https://vertabelo.com/blog/normalization-101/>

For: Normalization 101, including 1NF

2c. Second Normal Form (2NF)

<https://www.geeksforgeeks.org/dbms/second-normal-form-2nf/>

For: 2NF definition, rules, and examples

<https://vertabelo.com/blog/normalization-101/>

For: Normalization 101, including 2NF

2d. Third Normal Form (3NF)

<https://www.geeksforgeeks.org/dbms/third-normal-form-3nf/>

For: 3NF definition, rules, and examples

<https://vertabelo.com/blog/normalization-101/>

For: Normalization 101, including 3NF

2e. Avoiding data redundancy (incl. Denormalization — hidden concept)

<https://vertabelo.com/blog/normalization-101/>

For: Practical redundancy-avoidance techniques

<https://www.geeksforgeeks.org/dbms/normal-forms-in-dbms/>

For: Normal forms summary, redundancy elimination

Topic 3: SQL Joins

3a. INNER JOIN

<https://dev.mysql.com/doc/refman/8.0/en/join.html>

For: Official MySQL JOIN syntax reference

https://www.w3schools.com/sql/sql_join_inner.asp

For: INNER JOIN syntax and examples

3b-3e. LEFT/RIGHT/FULL JOIN, Multi-table queries

<https://dev.mysql.com/doc/refman/8.0/en/join.html>

For: Official MySQL JOIN syntax reference (all join types)

https://www.w3schools.com/sql/sql_join_left.asp

For: LEFT JOIN syntax and examples

https://www.w3schools.com/sql/sql_join_right.asp

For: RIGHT JOIN syntax and examples

https://www.w3schools.com/sql/sql_join_full.asp

For: FULL JOIN syntax and examples (concept; MySQL needs UNION workaround)

Topic 4: Advanced Querying

4a. Subqueries

<https://dev.mysql.com/doc/refman/8.0/en/subqueries.html>

For: Official MySQL subqueries reference

https://www.w3schools.com/mysql/mysql_subqueries.asp

For: Subquery syntax and examples

4b. Nested queries (correlated subqueries)

<https://dev.mysql.com/doc/refman/8.0/en/correlated-subqueries.html>

For: Official correlated subqueries reference

https://www.w3schools.com/mysql/mysql_subqueries.asp

For: Subquery syntax and examples

4c. Common Table Expressions (CTEs)

<https://dev.mysql.com/doc/refman/8.0/en/with.html>

For: Official WITH (CTE) reference

https://www.w3schools.com/sql/sql_cte.asp

For: CTE syntax and examples

4d. Complex filtering

<https://dev.mysql.com/doc/refman/8.0/en/case.html>

For: Official CASE operator reference

https://www.w3schools.com/sql/sql_case.asp

For: CASE WHEN syntax and examples

4e. Analytical queries

<https://mode.com/sql-tutorial/sql-analytics-training/>

For: SQL for analytics tutorial

<https://dev.mysql.com/doc/refman/8.0/en/aggregate-functions.html>

For: Official aggregate functions reference

Topic 5: Window Functions

5a. Ranking

<https://dev.mysql.com/doc/refman/8.0/en/window-function-descriptions.html>

For: Official window function descriptions (RANK, DENSE_RANK, ROW_NUMBER)

https://www.w3schools.com/sql/sql_ref_rank.asp

For: RANK() syntax and examples

5b. Partitioning

<https://dev.mysql.com/doc/refman/8.0/en/window-functions-usage.html>

For: Official PARTITION BY usage reference

https://www.w3schools.com/sql/sql_ref_partition_by.asp

For: PARTITION BY syntax and examples

5c. Running totals

<https://dev.mysql.com/doc/refman/8.0/en/window-functions-frames.html>

For: Official window frame specification reference

https://www.w3schools.com/sql/sql_ref_over.asp

For: Running totals with OVER() syntax

5d. Analytical calculations (LAG/LEAD)

<https://dev.mysql.com/doc/refman/8.0/en/window-function-descriptions.html>

For: Official LAG/LEAD function reference

https://www.w3schools.com/sql/sql_ref_lag.asp

For: LAG/LEAD syntax and examples

5e. Business reporting use cases

<https://mode.com/sql-tutorial/sql-window-functions/>

For: Window functions for reporting

<https://dev.mysql.com/doc/refman/8.0/en/window-functions.html>

For: Official window functions overview

Topic 6: Database Performance

6a. Indexes

<https://dev.mysql.com/doc/refman/8.0/en/mysql-indexes.html>

For: Official MySQL indexing reference

https://www.w3schools.com/sql/sql_create_index.asp

For: CREATE INDEX syntax and examples

6b. Query optimization

<https://dev.mysql.com/doc/refman/8.0/en/select-optimization.html>

For: Official SELECT optimization reference

<https://use-the-index-luke.com/>

For: In-depth SQL performance and indexing guide

6c. Execution awareness (EXPLAIN)

<https://dev.mysql.com/doc/refman/8.0/en/explain-output.html>

For: Official EXPLAIN output format reference

<https://use-the-index-luke.com/>

For: Understanding execution plans

6d. Efficient querying

<https://dev.mysql.com/doc/refman/8.0/en/select-optimization.html>

For: Official query optimization reference

<https://use-the-index-luke.com/>

For: Efficient SQL querying practices

6e. Scalable data retrieval

<https://dev.mysql.com/doc/refman/8.0/en/optimization.html>

For: Official MySQL scalability/optimization reference

<https://aws.amazon.com/products/databases/>

For: Database scalability best practices

Topic 7: Data Warehousing Concepts

7a. OLTP vs OLAP

<https://www.ibm.com/topics/oltp-vs-olap>

For: OLTP vs OLAP explained

<https://learn.microsoft.com/en-us/azure/architecture/data-guide/relational-data/online-analytical-processing>

For: OLAP concepts from Microsoft

7b. Data warehouses

<https://aws.amazon.com/data-warehouse/>

For: What a data warehouse is (AWS)

<https://cloud.google.com/learn/what-is-a-data-warehouse>

For: Data warehouse concepts (Google Cloud)

7c. Analytical databases

https://cloud.google.com/bigquery/docs/storage_overview

For: BigQuery's column-oriented storage architecture

<https://www.snowflake.com/en/data-cloud/architecture/>

For: Snowflake's columnar storage architecture

7d. Reporting systems

<https://www.tableau.com/learn/articles/business-intelligence>

For: What business intelligence/reporting is

<https://learn.microsoft.com/en-us/power-bi/fundamentals/power-bi-overview>

For: Power BI reporting concepts

7e. ML data pipelines

<https://cloud.google.com/architecture/data-preprocessing-for-ml-with-tf-transform-pt1>

For: ML data pipeline preprocessing guide

<https://www.databricks.com/glossary/feature-engineering>

For: Feature engineering for ML

Note: some sources repeat across subtopics because a single reference page (e.g. a full MySQL documentation page) covers several related commands or concepts at once.